

Sanjana Shree Bangari

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PROFESSIONAL SUMMARY

Data Engineer with 3+ years of experience in building scalable ETL pipelines, data modeling, and warehousing using cloud services like AWS. Proficient in SQL, Python, and big data technologies such as da, Hive, Spark, and EMR to process and extract insights from large Mtasets. **Strong foundation with 4 years of experience in Data Structures, Algorithms, and Object-Oriented Programming. AI/ML enthusiast with hands-on experience in machine learning model development, fine-tuning large language models, and deploying scalable ML solutions.**

EDUCATION

Purdue University, Indianapolis

Jan 2023- Dec 2024

Applied Data Management and Analytics / Master of Science GPA: 3.86/4.0

Visvesvaraya Technological University

Computer Science and Engineering / Bachelor of Technology GPA: 4.0 / 4.0

Aug 2018 – July 2022

Coursework: Software Engineering, Machine Learning, Statistics, Deep Learning, Data Mining, Big Data, Natural Language Processing, Artificial Intelligence and Data Structures

WORK EXPERIENCE

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ML Engineer

Feb 2025 – July 2025

- Designed a graph-based predictive intelligence system using Neo4j, reducing network fault detection latency by 37% and improving outage response rates by 45%.
- Modeled over 1.2 million network entities and applied graph algorithms (FastRP, node2vec, Louvain) to enhance churn prediction accuracy by 24% and data retrieval speed by 60%.
- Developed a low-latency anomaly detection pipeline with Python and XGBoost, improving fault prediction precision by 33% and reducing false positives by 19%.
- Integrated LLMs (GPT-4, Cohere) through a RAG framework to generate real-time diagnostics from Neo4j graphs and network logs, automating 48% of manual triage tasks.
- Collaborated cross-functionally to productionize ML models and deploy predictive alerts, decreasing SLA violations by 22% and accelerating incident resolution by 40%.

Purdue University

AI/ML Research Assistant

Jan 2023 - Dec 2024

- Led cutting-edge research in NLP and Federated Learning by fine-tuning Large Language Models (LLMs, 7B+ parameters) using Low-Rank Adaptation (LoRA), resulting in a 25% increase in training efficiency and 50% reduction in computing overhead across distributed systems.
- Engineered resilient data pipelines for anomaly detection in federated training workflows, ensuring real-time filtering of compromised client models and improving model integrity by 30%, aligning with security-first design principles.
- Designed and simulated data poisoning attacks on federated setups to evaluate and enhance defense mechanisms like Hawkeye, demonstrating a 40% increase in detection accuracy through controlled adversarial testing and metrics-based evaluation.
- Automated model orchestration and training pipelines using Python and shell scripting, facilitating seamless scalability across academic compute nodes and reducing deployment latency by 35%.

GTechnologies Pty Ltd

Software Development Engineer

Jan 2023- Dec 2024

- Built and deployed a responsive office notice board app using MERN stack, reducing internal communication latency by 45% and improving update visibility by 72%.
- Designed secure RESTful APIs with JWT authentication and RBAC, ensuring 100% protection against unauthorized access and enabling efficient CRUD operations.
- Optimized MongoDB schemas with indexing and validation, leading to 28% faster data retrieval and scalable storage of employee, meeting, and announcement data.
- Integrated WebSockets for real-time updates, eliminating refresh-based polling and cutting message delivery latency by 45%.
- Deployed on AWS with CI/CD pipelines and CloudWatch monitoring, improving deployment agility by 65% and reducing bug resolution time by 30%.

Gustovalley Technovations

- Led the development of scalable data pipelines using Python and R to automate data ingestion, processing, and transformation of large datasets, increasing pipeline efficiency by 40% and reducing manual effort by 70%.
- Managed databases using SSMS and wrote optimized SQL queries to ensure high-integrity storage and fast retrieval, reducing query latency by 35% and improving data accuracy.
- Designed and orchestrated cloud-based data integration workflows using AWS EMR and Azure, achieving 99.9% uptime and lowering data transfer latency by 30%.
- Built robust ETL pipelines using SQL and Python, aligned with business logic and governance standards, cutting data redundancy by 45% and enabling 20% faster onboarding of new data sources.
- Created dynamic dashboards and reports using Power BI and SSRS, empowering stakeholders with visibility into KPIs and enabling 20% faster decision-making.
- Utilized Python libraries (Pandas, NumPy, Scikit-learn, Matplotlib) for data manipulation, modeling, and visualization to support machine learning and analytics use cases.
- Developed scalable data models and implemented data warehousing solutions to support enterprise analytics and reporting, improving reporting efficiency by 25%.
- Executed big data processing using Hive, Pig, Apache Spark, Spark Streaming, and AWS EMR to derive insights from high-volume, real-time datasets, reducing analysis time by 40%.
- Applied MapReduce algorithms for large-scale data pattern detection, supporting strategic business decisions through large dataset analysis.
- Improved performance of PySpark jobs by optimizing transformation logic, tuning Spark configurations, and reducing execution time by 50%.

SKILLS

Languages: Java, Python, C++, JavaScript, TypeScript, Scala, KornShell, Go

Frameworks & Tools: Spring Boot, React, Node.js, JUnit, Mockito, Git, Jenkins, Maven, FastAPI, Streamlit, LangChain, Airflow, MLflow, Apache NiFi, Talend, Informatica, Vue, GraphQL

Machine Learning & AI: TensorFlow, PyTorch, Scikit-learn, Hugging Face, LLMs (GPT, BERT, T5), RAG, LoRA, PEFT, Prompt Engineering, Computer Vision, ML Pipelines, Model Evaluation, Optimization, Alteryx, NLP, Neural Networks, JAX

Agentic AI: LangGraph, CrewAI, AutoGen, Multi-Agent Systems, Tool-Using Agents, Memory-Augmented Reasoning

Big Data & Analytics: Hadoop, Hive, PySpark, Spark, Kafka

Cloud Platforms: AWS (Lambda, S3, EC2, SQS, SageMaker, CloudWatch, Bedrock, Glue, Redshift, EMR), Azure (Data Factory, Synapse Analytics, Databricks), Docker, Kubernetes, Terraform, Datalakes, BigQuery

Databases: PostgreSQL, MySQL, MongoDB, Cassandra, Redis, Neo4j, Pinecone, FAISS, Snowflake

Development Practices: Agile Methodologies, CI/CD Pipelines, RESTful API Development, Test-Driven Development, Microservices, Serverless

Data Visualization: PowerBI, Excel, Tableau

Core Strengths: Software Design, System Design, Data Structures & Algorithms, Debugging, Problem Solving, Collaboration, Version Control

ACHIEVEMENTS

IBM Data Science Professional Certificate | Google Data Analytics Professional Certificate | Presented Project titled 'CONCEALED WEAPON DETECTION USING IMAGE FUSION' in National Conference, INDIA | Participated in Infosys HackWithInfy, TCS CodeVita, Meta HackerCup | Organized 5+ University Level Coding Contests |

PROJECTS

Autonomous Research Assistant using Agentic AI: Built a multi-agent system using LangGraph and CrewAI where agents collaboratively performed literature review, extracted insights, and generated structured reports from research papers.

Snowflake Data Warehouse Pipeline: Built a Snowflake-based data warehouse with Python ETL pipelines, improving data processing efficiency by **30%**. Optimized SQL queries and stored procedures, reducing query time by **40%**, and integrated Tableau dashboards that enhanced business reporting accuracy by **25%**.

Customer Churn Prediction & BI Dashboard: Developed a churn prediction model using AWS SageMaker, built SQL pipelines for data processing, and created **Tableau dashboards** to visualize churn risk and customer insights.

Data Pipeline Automation with Airflow & Docker: Designed and deployed a containerized data pipeline on Azure using Airflow and Terraform, integrating CI/CD with GitHub Actions for automated testing and deployment.